

Jonathan Barnes | Applied Data Scientist

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PROFESSIONAL EXPERIENCE

University of Minnesota, Minneapolis, MN

Analytics Engineer | Center for Public Health Systems

(May 2025 - Present)

- Established and operate an in-house analytics and data warehousing platform for federal health funding analysis.
- Integrated and standardized 3M+ HHS award records into a unified analytical model supporting evaluation.
- Designed analytical data models to reconcile complex federal funding and organizational structures.
- Architected modular, logged data pipelines and data modeling layers, enabling scalable, reproducible analytics.
- Authored complex Python and SQL queries over large-scale funding data for analytical workflows.
- Delivered analysis-ready datasets and Tableau dashboards communicating HHS funding trends, allocations, and flows.
- Enabled program-level evaluation of PHIG funding allocations using standardized federal award data.
- Developed internal tooling to support controlled execution and refresh of analytical data assets.

Graduate Teaching Assistant | Division of Biostatistics and Health Data Science

(Aug 2024 - May 2025)

- Taught and evaluated applied statistical analysis in R, improving student code quality and statistical reasoning.

Graduate Research Assistant | Medical School

(Oct 2023 - May 2024)

- Designed custom plasma biomarker phenotypes and gene expression features to enable survival modeling.
- Applied Cox proportional hazards regression to model time to cognitive decline using biomarker phenotypes.

University of St. Thomas, St. Paul and Minneapolis, MN

ITS Support Technician Intern & Senior Student Technician | ITS Support Services

(Aug 2021 - Aug 2023)

- Resolved or consulted on 4,500+ IT support tickets across hardware, software, networking, and security.
- Developed a Python application that enabled core ITS service desk workflows and improved operational efficiency.
- Mentored junior student technicians to independently handle complex service desk operations and consultations.
- Imaged and administered software on 1,500+ summer replacement computers, doubling the typical annual volume.
- Assembled an in-house virtual lab to replace AWS EC2 instances, reducing cloud footprint and infrastructure costs.
- Implemented digital signage across service desks and campuses to improve access to real-time IT service information.
- ITS Outstanding Student Award.

EDUCATION

University of Minnesota School of Public Health, Minneapolis, MN

Master of Public Health - Public Health Data Science, GPA 3.73/4.00

(Expected Jan 2026)

- APEx — Pre-Statistics Modules for MPH Students
- ILE — Evaluation of US Federal Spending and Systems.
- President's Student Leadership & Service Award Nominee (2025)

University of St. Thomas, St. Paul, MN

Bachelor of Science - Data Analytics (Biology Domain), Completed while employed full-time

- Minors - Biology (Biochemistry & Computational Biology courses), Applied Statistics.
- Capstone - Development of Obesity and Nutritional Trends Over Time in the United States Using Combined NHANES.

SKILLS

Languages: R, Python, SQL, Bash.

Tools: Shiny, APIs, Git, ArcGIS, Tableau, CI/CD, Microsoft Azure, Docker.

Competencies: Data Visualization, Data Communication, ETL Pipelines, Geospatial Visualization, Longitudinal Data Analysis, Machine Learning, Artificial Intelligence, Relational Databases, Columnar Data Architectures, System Design, Experimental Design, Data Acquisition, Consulting, Health Informatics, Electronic Health Records, HIPAA, PHI, SNOMED, ICD-10.

Statistical Methods: Dimensionality Reduction, Supervised Learning, Unsupervised Learning, Regression, Regularization, Tree Models, Mixed Effects Models, Survival Analysis, Structural Equation Modeling, Natural Language Processing.

PROJECTS

Public Health Federal Funding Evaluation Platform & Dashboard (Grant Supported)

- Developed a modular ingestion and modeling pipeline to handle millions of U.S. federal public health funding records, spanning prime awards, sub-awards, and derived transactions across inconsistent reporting structures.
- Designed reconciliation logic to accurately link awards, recipients, and health departments across divergent reporting hierarchies, identifier schemes, and funding relationships.
- Implemented rules-based normalization to standardize agencies, recipients, locations, and award classifications across fragmented HHS source systems with inconsistent metadata.
- Resolved duplicated entities, missing identifiers, and conflicting award relationships in large-scale federal data.
- Optimized storage using DuckDB and Parquet, reducing data size by 75%+ while preserving query performance.
- Materialized analysis-ready datasets and Tableau Hyper extracts for reproducible visualization workflows.
- Built a GUI-based orchestration tool to manage pipeline execution, refresh cycles, and validation.
- Designed dashboards to analyze funding flows, geographic distribution, and program-level spending.

Methods & Tools: Python, Polars, SQL, DuckDB, Apache Parquet, Tableau (Hyper), Data Modeling, Schema Design, Entity Resolution, Data Normalization, Data Pipelines, Data Warehousing, Data Validation, Health Informatics.

Tech Desk Tools Python

- Designed and deployed a user-facing Python application that serves as the primary interface for Tech Desk workflows, enabling technicians to initiate identity lookup, diagnostics, and institutional tool access through a unified system.
- Implemented identity derivation and system automation logic that abstracts OS and directory complexity into stable, technician-ready outputs, minimizing manual troubleshooting and workflow variability.
- Integrated OS-level commands and Active Directory queries within an Azure-backed identity environment, with built-in error handling and privilege-aware logic to ensure consistent behavior across diverse account and system states.
- Implemented role-specific administrative features for technicians based on directory permissions and privileges.
- Adopted as the standard Tech Desk application, supporting continuous daily operations since early 2023.

Methods & Tools: Python, PowerShell, Bash, App Development, Active Directory, Azure, Systems Automation, Workflow Orchestration, Error Handling & Defensive Programming, Technical Writing, Packaging and Deployment.

Mapping Misery – Predicting Poor Physical and Mental Health Days in the U.S.

- Developed predictive models for poor physical and mental health days in 3,000+ U.S. counties using R.
- Engineered clusters based on 10 years of trends across 15 metrics using Gaussian Mixture Models (GMM) by transforming variables into 8 level categories with NAs for robust Multiple Correspondence Analysis (MCA).
- Achieved an R^2 of 0.844 (RMSE: 0.263) for physical health days with Elastic Net Regression.

Methods & Tools: Data Pipelines, PCA, MCA, GMM Clustering, Feature Engineering, Longitudinal Data Analysis, Elastic Net Regression, Predictive Modeling, R, Geospatial Visualization, GIS, Data Visualization.

Applied Biostatistics Introduction Modules (Grant Supported)

- Designed applied biostatistics modules translating probability, inference, and uncertainty into intuitive frameworks while embedding ethical interpretation, contextual reasoning, and bias awareness in public health analysis.

Methods & Tools: Biostatistics, Statistical Reasoning, Probability, Curriculum Design, Technical Writing, R.

Data Inspection Dashboard | <https://jonathanbarnes.shinyapps.io/DescriptiveApp/>

- Developed an interactive data inspection tool that automates EDA, data validation, missingness handling, imputation, and statistical summaries for common public health datasets.

Methods & Tools: R, Shiny, Data Pipelines, Dashboards, Data Visualization, Data Communication, JavaScript.

Predicting Child Mortality and Cesarean Delivery Outcomes with Generalized Linear Mixed Models.

- Built Generalized Linear Mixed Models (GLMM) to investigate factors affecting child mortality and cesarean delivery.
- Determined a 1,124.5% increase in the odds of cesarean delivery for hospital births (95% CI: 618.7% - 1,986.3%, $p < 0.001$) — a result consistent with logical expectations.

Skills & Tools: GLMM, GEE, R, Data Cleaning, Data Pipelines, Data Visualization, Feature Engineering, Collaboration.